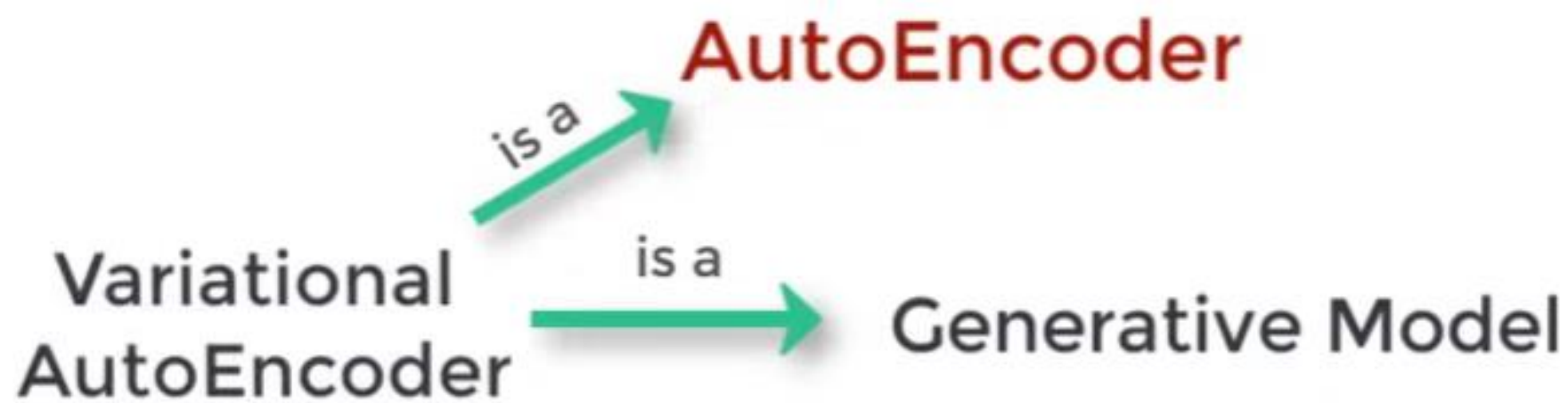
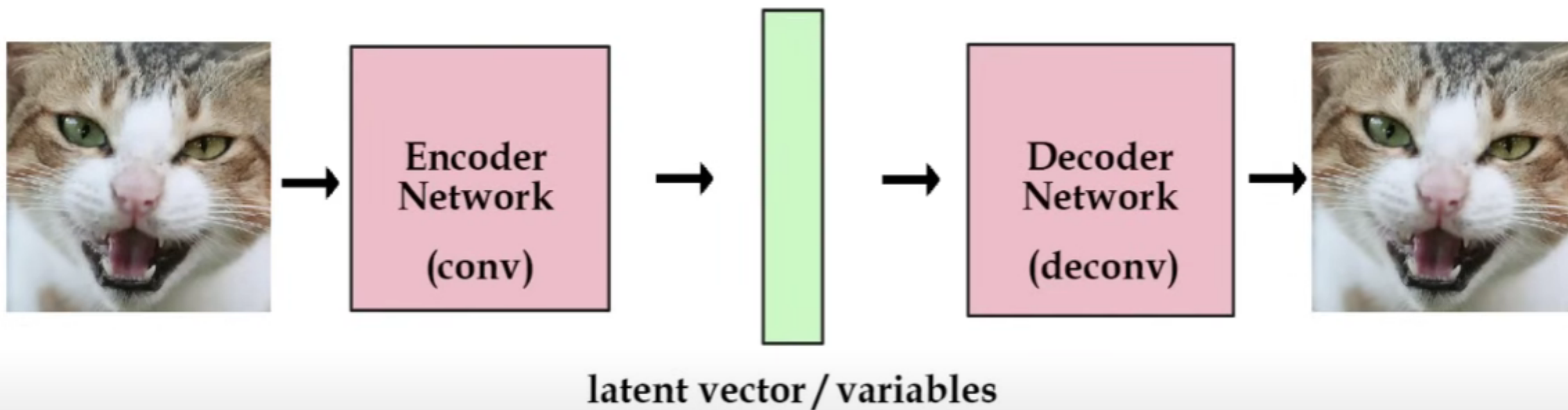
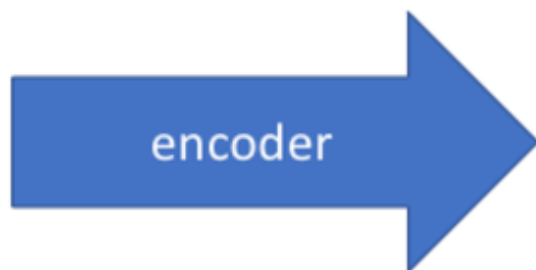


Variational Autoencoder

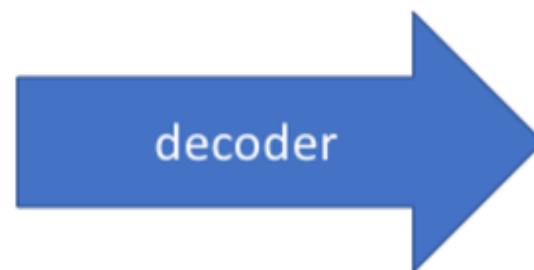


AutoEncoder





Latent attributes

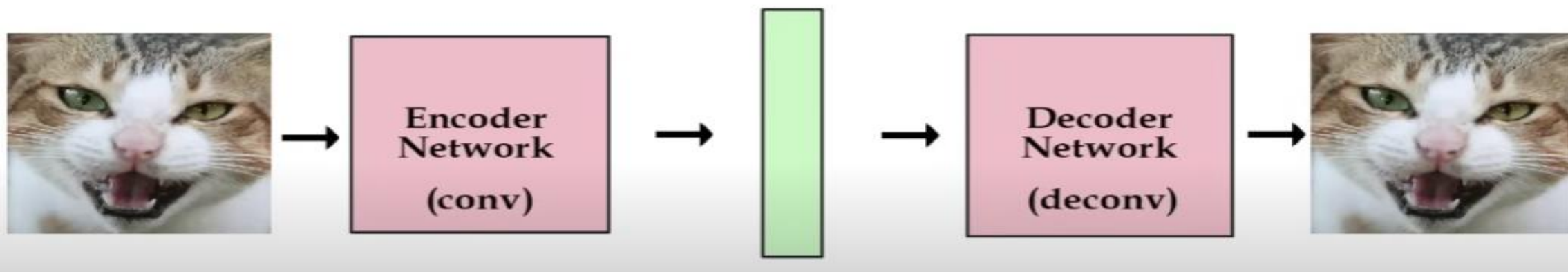


AutoEncoder

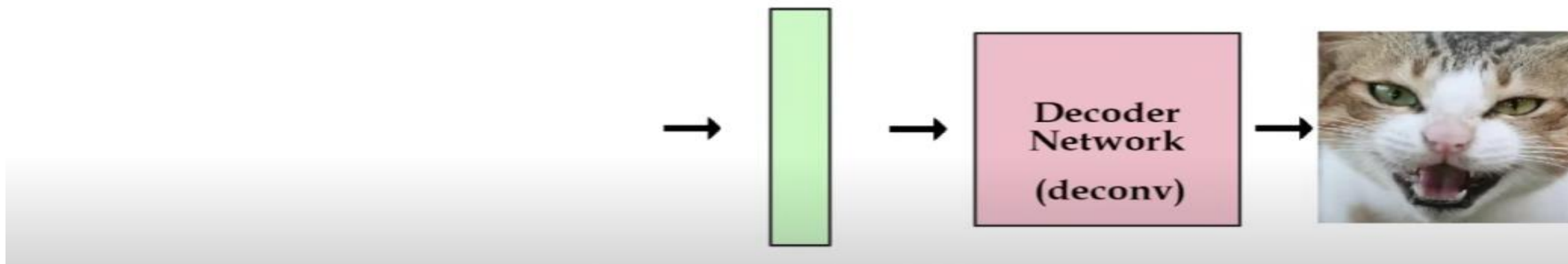
AutoEncoder



Training Phase



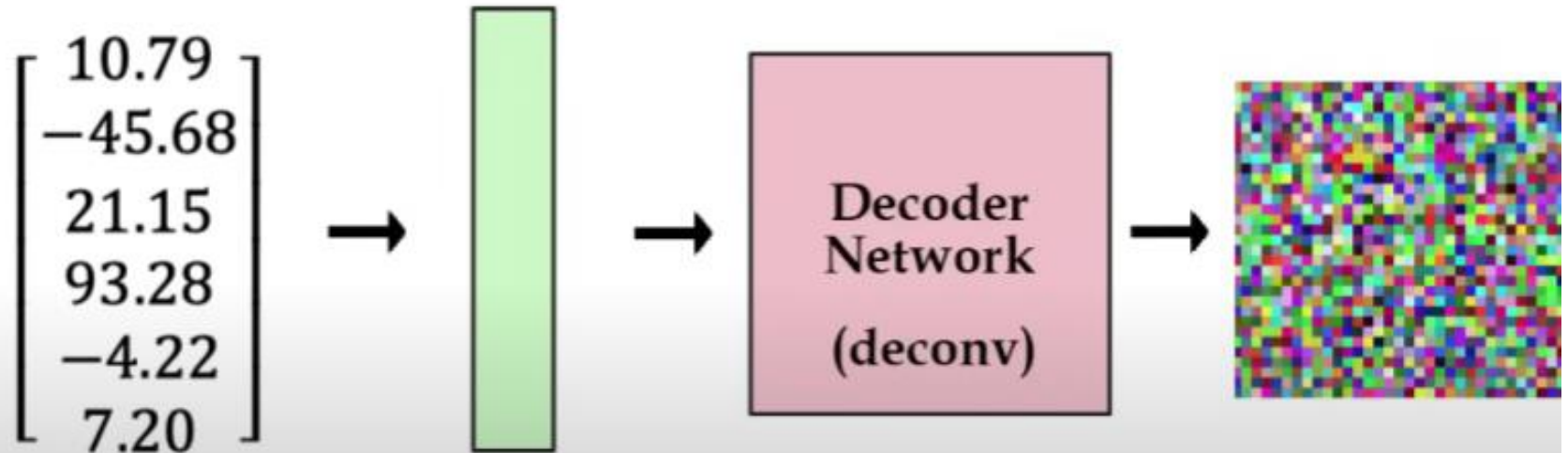
Testing Phase



AutoEncoder



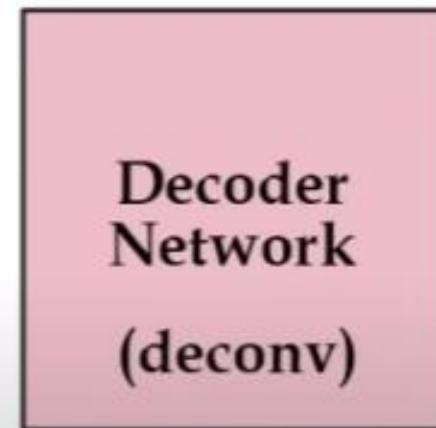
Testing Phase



AutoEncoder

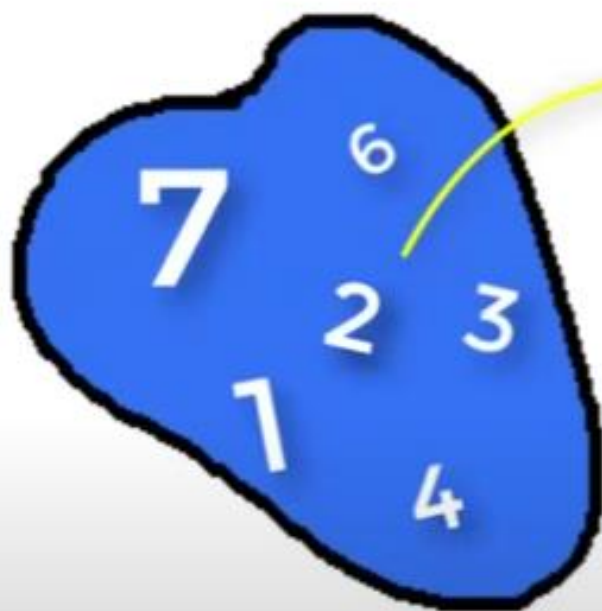
Testing Phase

“Sample” from
a “Distribution”



latent vector / variables

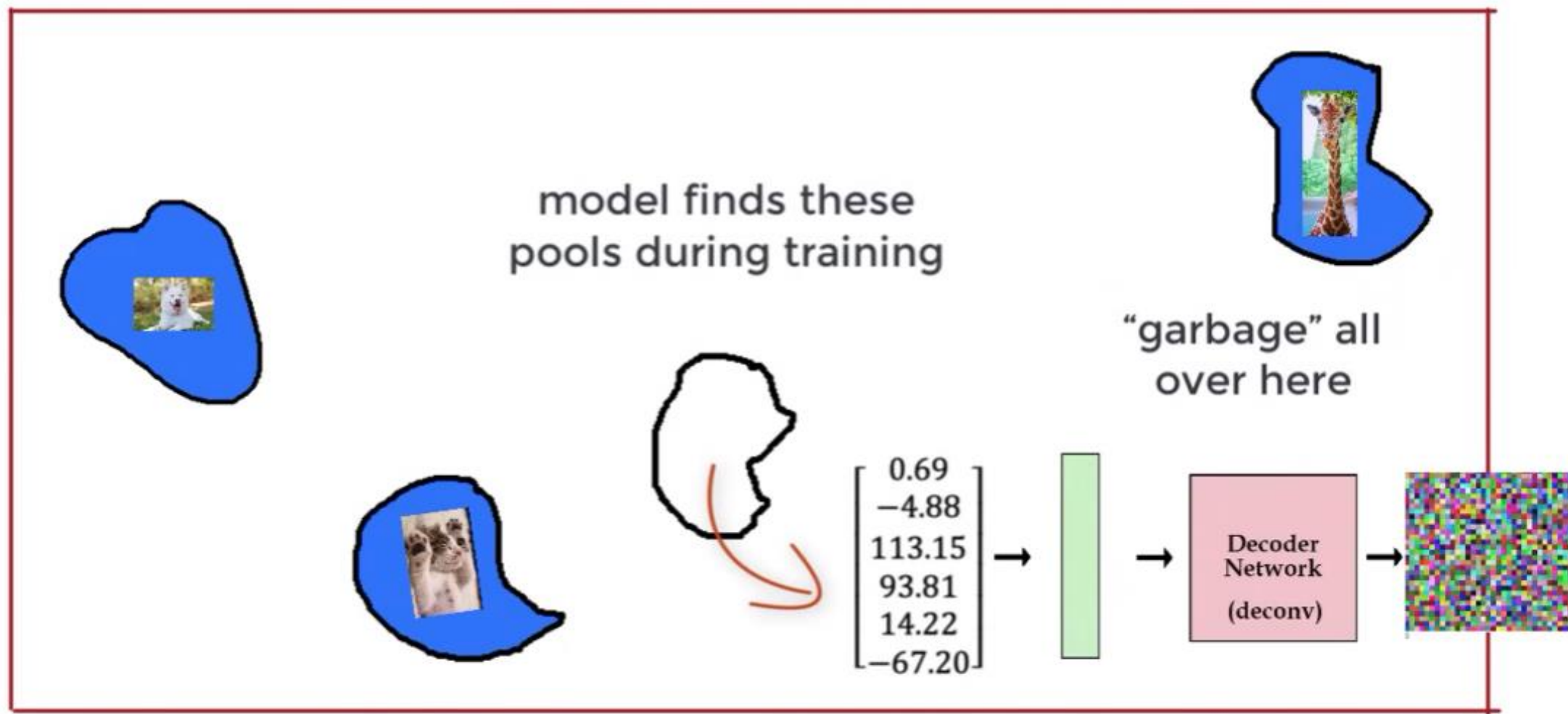
Distribution - Concept


$$\begin{bmatrix} 0.69 \\ -4.88 \\ 113.15 \\ 93.81 \\ 14.22 \\ -67.20 \end{bmatrix}$$

Distribution

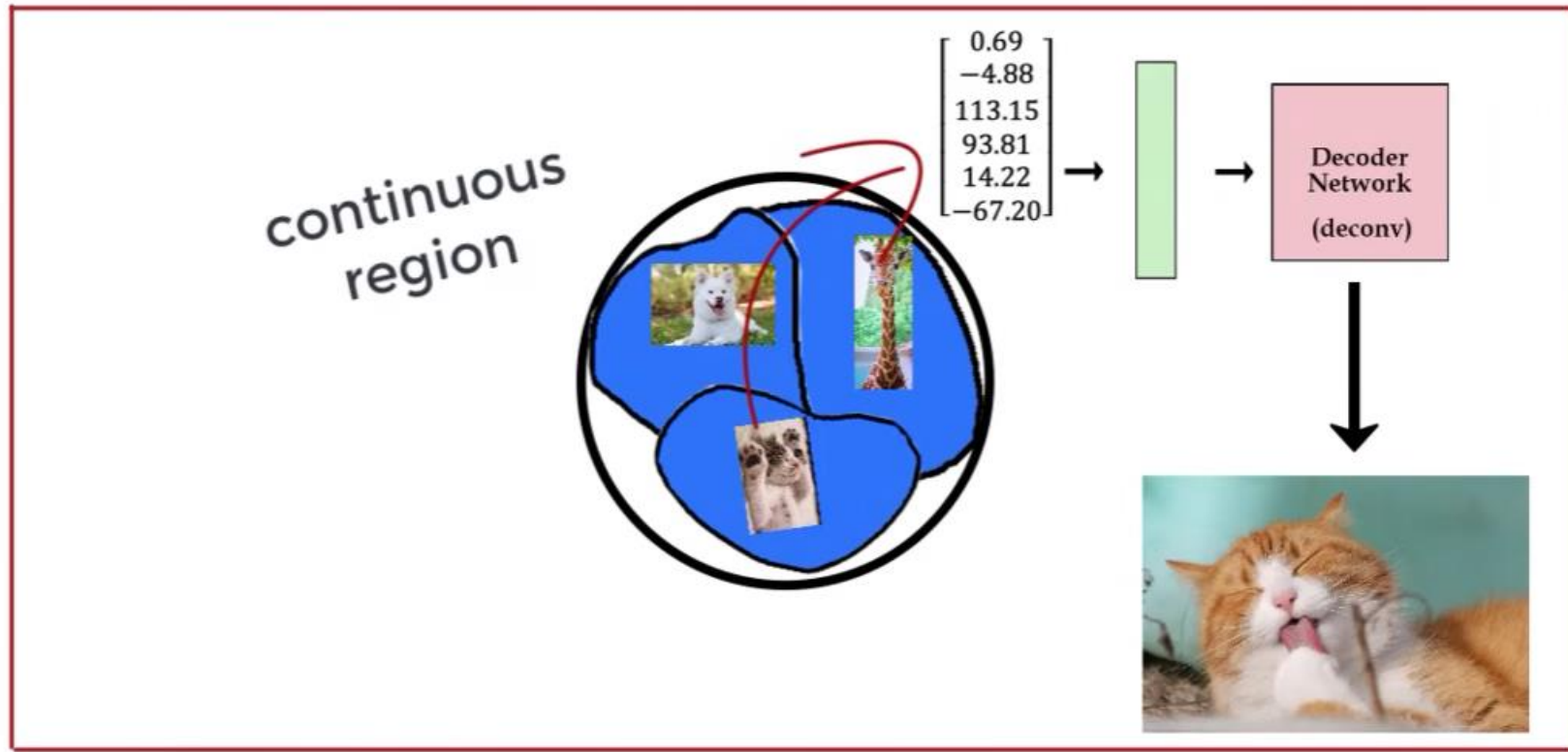


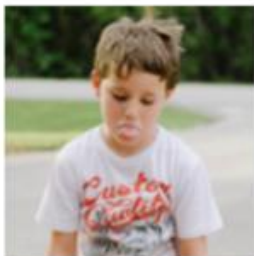
Sampling - Concept



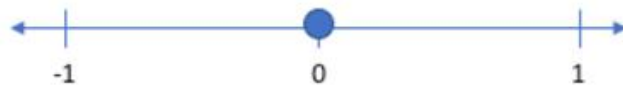
Variational AutoEncoder

Sampling - Concept

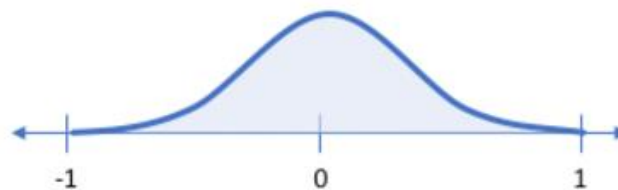
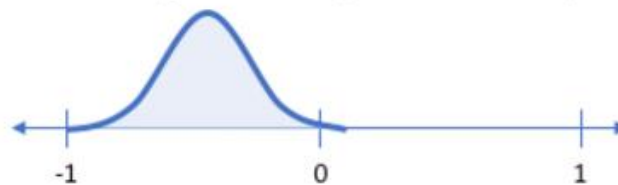




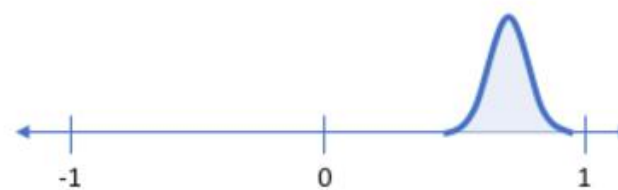
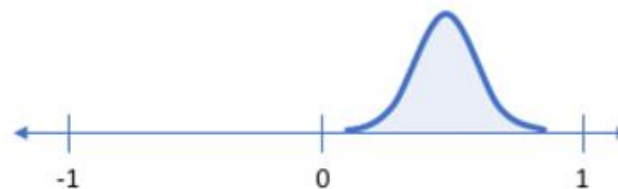
Smile (discrete value)



Smile (probability distribution)

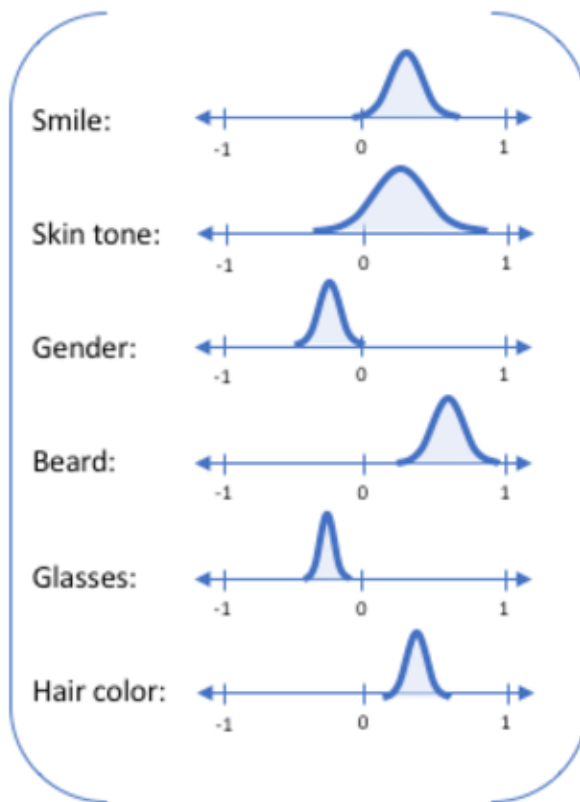


vs.



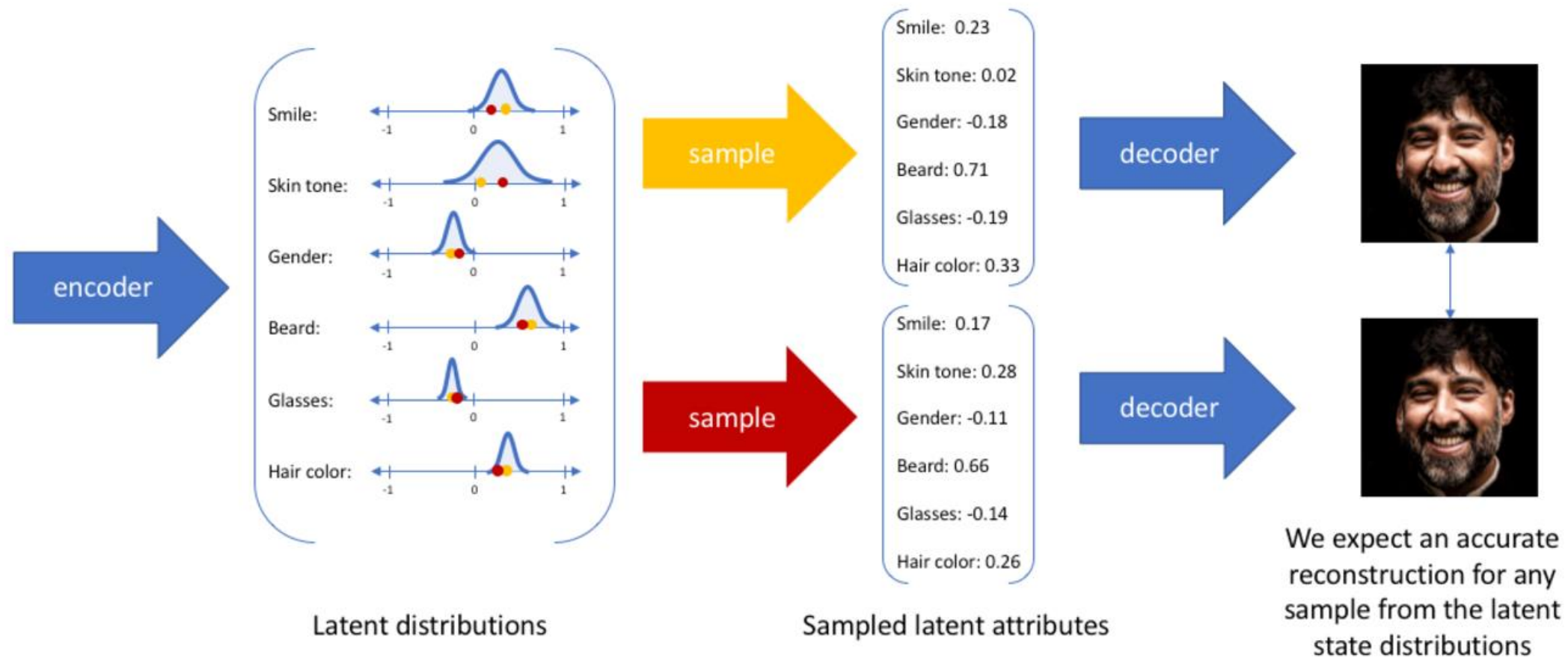


encoder



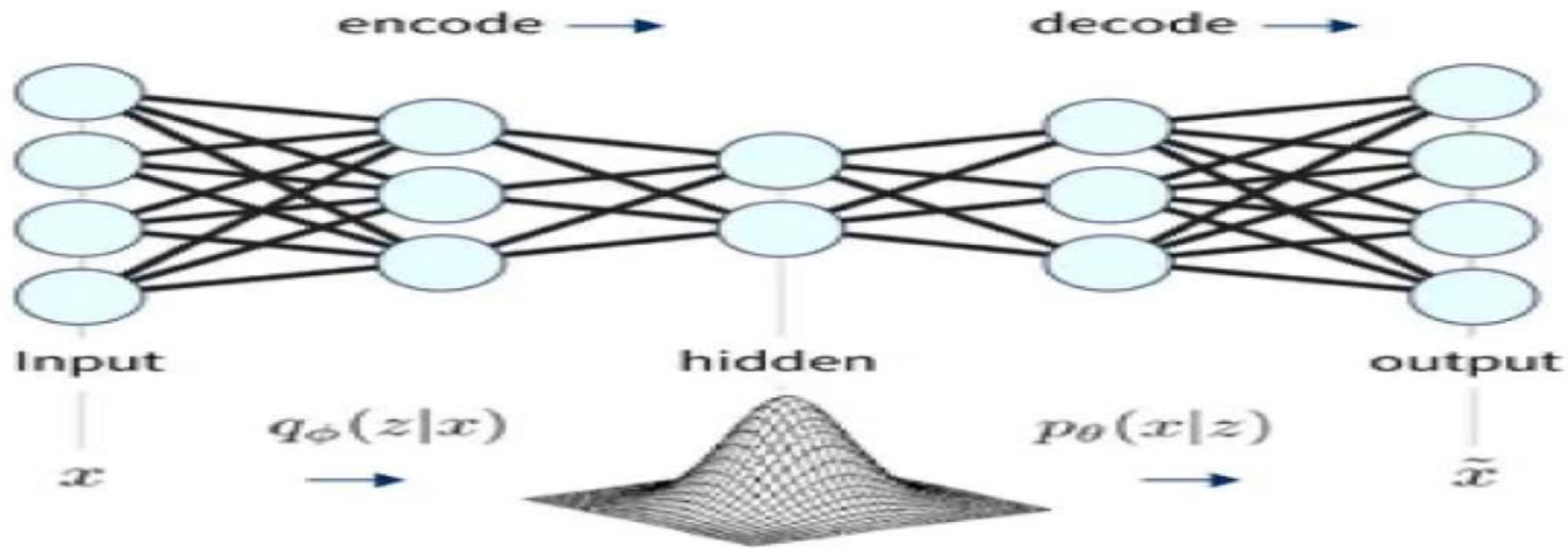
decoder

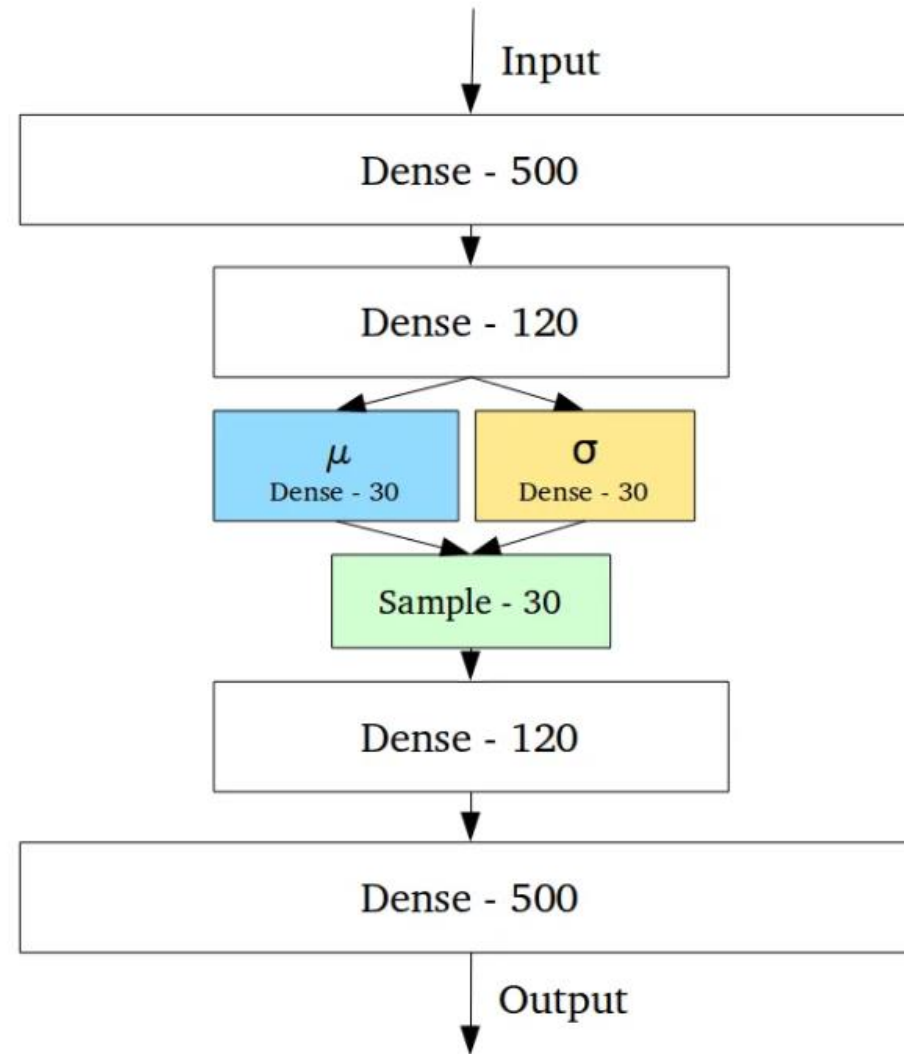




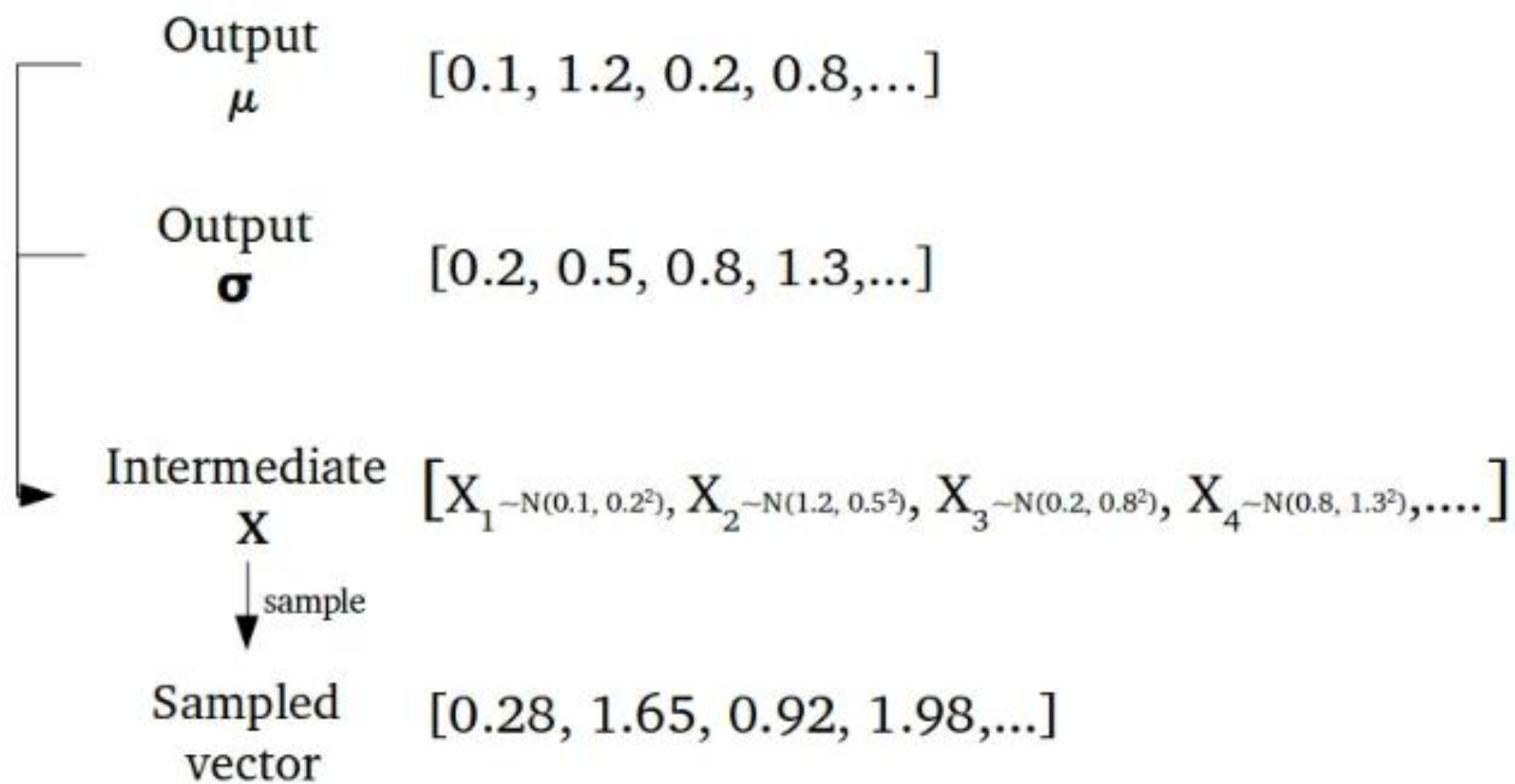
Here is how the process flow looks:

1. The input data x is fed into the encoder, which outputs the parameters of the latent space distribution $q(z/x)$ (mean μ and variance σ^2).
2. Latent variables z are sampled from the distribution $q(z/x)$ using techniques like the reparameterization trick.
3. The sampled z is passed through the decoder to produce the reconstructed data $\hat{x}$, which should be similar to the original input x .

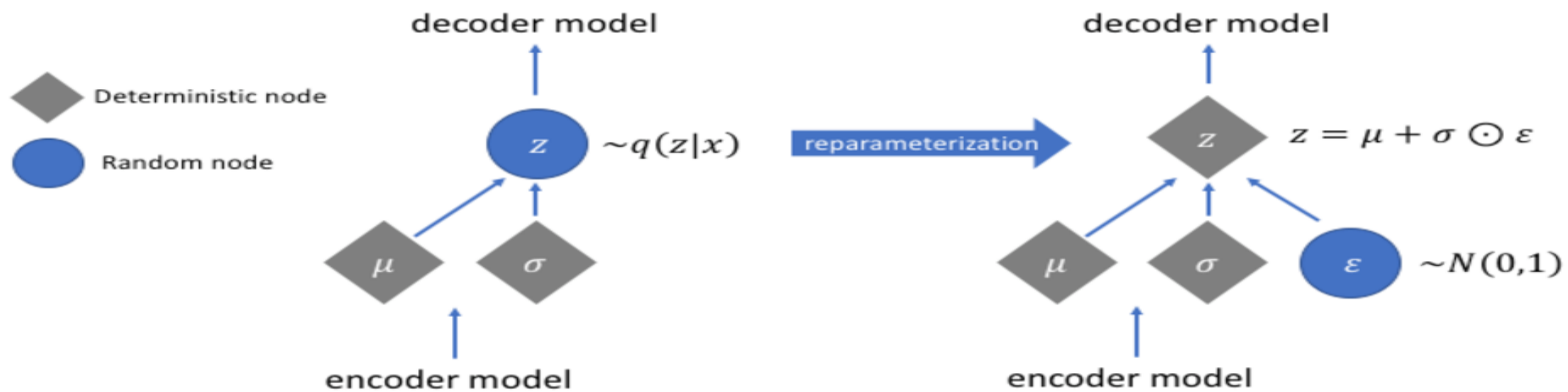
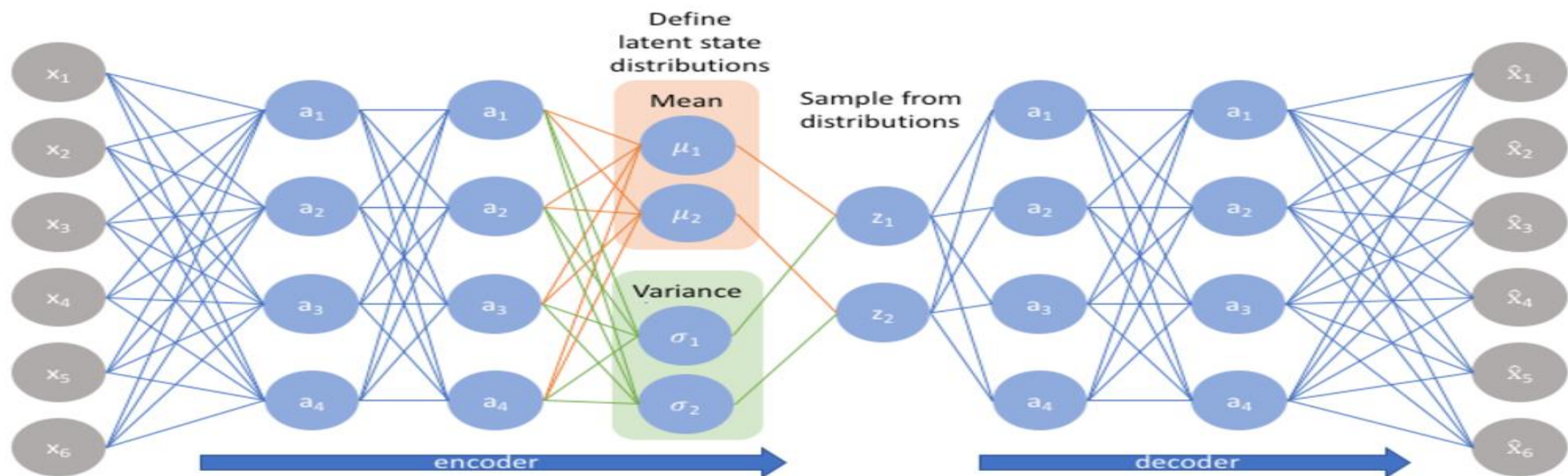


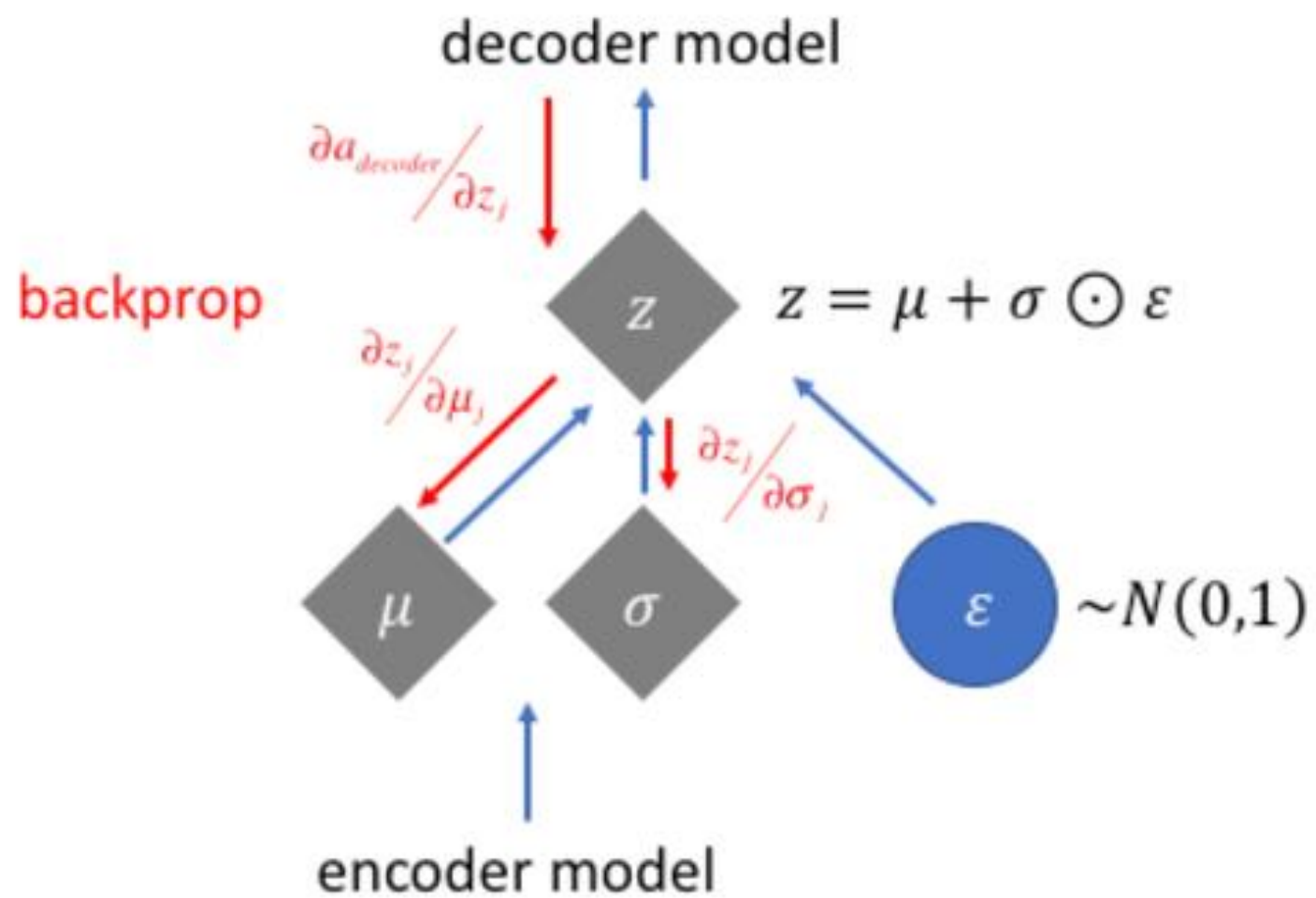


Variational Autoencoder



Stochastically generating encoding vectors





Reparameterized trick

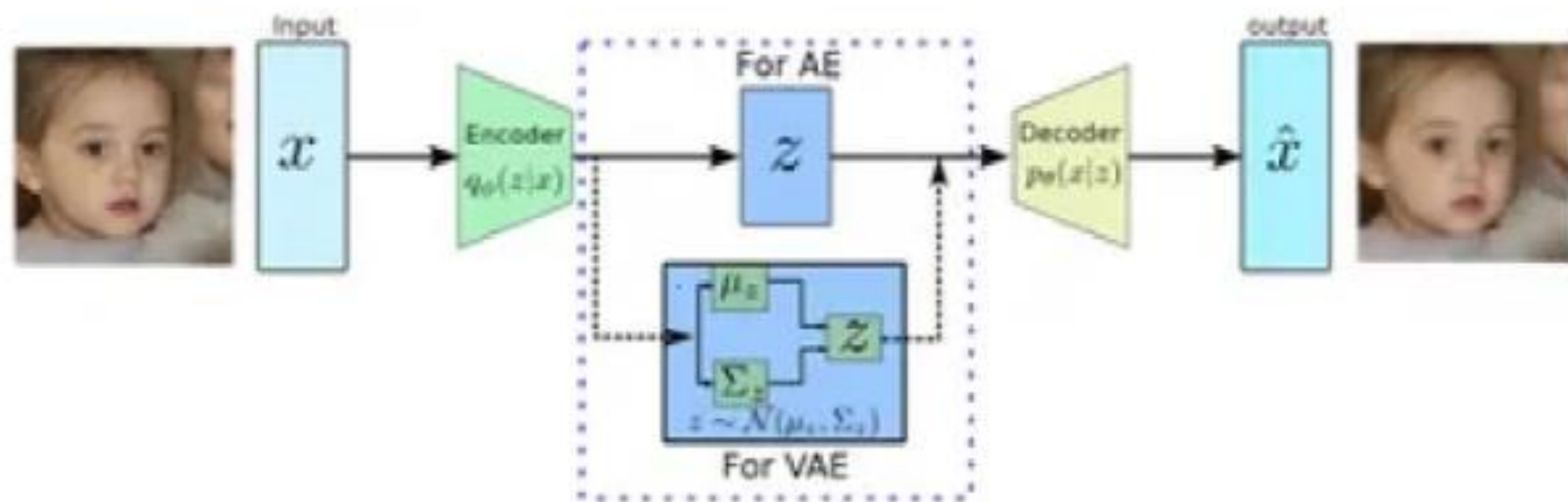
1. **Introduce a new random variable:** We introduce another independent random variable, typically another standard normal variable (ϵ). This new variable (ϵ) serves as a source of randomness entirely separate from the sampling process within the VAE.
2. **Reparameterize the latent variable:** Instead of directly adding noise, we create a scaled version of the new random variable (ϵ) using the standard deviation (σ) predicted by the encoder. We achieve this with element-wise multiplication:

$$z = \mu + \sigma * \epsilon$$

This equation might seem very similar to the original equation, but there's a crucial difference. Now, both μ and σ are deterministic outputs from the encoder, and ϵ is a separate source of randomness we can control.

Benefits:

- **Backpropagation is Happy:** Because the new equation only involves deterministic operations, we can now calculate gradients with respect to μ and σ during backpropagation. This allows the network to learn and adjust its parameters effectively.
- **Preserves Stochasticity:** Even though the reparameterization trick uses a deterministic transformation, the final sampled latent variable (z) retains the desired statistical properties of the original Gaussian distribution. This ensures the VAE maintains its ability to explore the latent space effectively.



Normal AutoEncoder

Why does this exist?

- Learn a hidden representation of input (that “vector”)
- Can NOT generate new data

What does this optimize?

- Minimize reconstruction loss

Variational AutoEncoder

- Learns to generate new data

- Minimize reconstruction loss + latent loss
- Latent vectors are sampled from Gaussian Mixture

Reference

- <https://www.jeremyjordan.me/variational-autoencoders/>
- <https://www.youtube.com/watch?v=fcvYpzHmhvA&t=757s>
- <https://towardsdatascience.com/intuitively-understanding-variational-autoencoders-1bfe67eb5daf>
- <https://www.analyticsvidhya.com/blog/2023/07/an-overview-of-variational-autoencoders/>